Evidence-Gated Claim Ledgers: A Reporting Standard for Pathology Foundation Models

Enrique Zueco, David Scott Lewis, Haley Yi
AIXC Research
reports@aixcresearch.com

Abstract

Pathology foundation models underwrite clinical-adjacent decisions across grading, subtype prediction, and biomarker inference, yet the field has no shared surface for recording what a model asserted about an individual slide and what verdict a common standard would assign to that assertion. Current practice reports aggregate metrics and cites confidence scores that carry no shared operational meaning across backbones. We introduce EGCL — Evidence-Gated Claim Ledgers — a lightweight, model-agnostic reporting standard for per-slide pathology-FM outputs. EGCL treats every prediction as a typed `Claim`, routes it through an `Evidence Gate` that returns a `Verdict` from a four-value vocabulary (SUPPORTED, UNSUPPORTED, NOT_APPLICABLE, CONTESTED), and persists the outcome to an append-only `Claim Ledger`. An `Anti-Fabrication Guard` refuses to release any downstream artefact whose numeric claims cannot be resolved to a SUPPORTED ledger entry or a declared ledger aggregation. On the public PANDA prostate cohort with the Phikon-v2 backbone we demonstrate an existence proof: on an $N{=}500$ cohort evaluated out-of-fold, the verdict vocabulary separates operationally. Under attention pooling (ABMIL) the gate releases 90 slides at SUPPORTED accuracy 0.811, whose 95% Wilson interval $[0.718, 0.879]$ does not overlap CONTESTED $[0.412, 0.514]$; a deliberately conservative mean-pool head releases only 9 slides (coverage 1.8%, accuracy 0.778) and is reported as the low-coverage floor. Both orderings run SUPPORTED $>$ CONTESTED $>$ the head's own balanced-accuracy base rate $>$ UNSUPPORTED — the ordering a calibrated model should produce, which is the sanity check the gate must pass rather than a gain it creates. The gate is stable across three MIL heads (mean-pool, ABMIL, CLAM-SB). The standard also surfaces a clinically-critical blind spot aggregate metrics hide: the SUPPORTED channel is class-monopolised by benign predictions under both heads — all 9 mean-pool releases and 83 of 90 under ABMIL — so it releases 2 and 15 missed cancers respectively.

1 Introduction

Pathology foundation models (pathology-FMs) have moved from technical benchmarks into pipelines that inform clinical-adjacent decisions: Gleason grading, tumour-margin assessment, biomarker inference from H&E slides, and cohort triage for downstream molecular assays. The community has responded with an ecosystem of pretraining scales [Filiot et al., 2023, Chen et al., 2024, Xu et al., 2024, Vorontsov et al., 2024, Wang et al., 2022], evaluation benchmarks, and interpretability probes.

What is missing is the piece that would let a reporting pathologist, a hospital governance board, or a regulator trust an *individual* prediction. When a downstream reader wants to know whether a particular per-slide output is trustworthy, there is no ledger to consult, no verdict vocabulary, and no way to distinguish a supported claim from a plausible-looking guess. Aggregate metrics on held-out

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cohorts and per-slide confidence scores answer neither question: the first summarises behaviour across slides one no longer has access to, and the second is model-specific, uncalibrated across backbones, and treated inconsistently by downstream systems [Kompa et al., 2021].

Selective classification [Chow, 1970, Cortes et al., 2016, Geifman and El-Yaniv, 2017, 2019] and conformal prediction [Vovk et al., 2005, Angelopoulos and Bates, 2021, Romano et al., 2020, Olsson et al., 2022] give the model half of what a reporting standard needs — a mechanism to abstain — but do not themselves standardise the vocabulary, the provenance surface, or the build-time enforceability needed for interoperable reporting across sites, backbones, and downstream systems. Reporting-quality checklists in adjacent fields — CONSORT-AI [Liu et al., 2020], TRIPOD+AI [Collins et al., 2024], MI-CLAIM [Norgeot et al., 2020], DECIDE-AI [Vasey et al., 2022] — structure what is reported at the *study* level but do not touch the per-prediction release surface.

Contribution. We introduce EGCL, a reporting standard sitting under all of these. EGCL is not proposed as a novel classifier or calibrator; it is a shared vocabulary, a provenance surface, and a build-time guard. Its four-value verdict vocabulary is small enough to standardise across sites, expressive enough to distinguish an honest "don't know" from an honest "no", and pluggable enough to admit new gate rules (confidence-threshold, entropy, margin, conformal) without changing the interface. To our knowledge, EGCL is the first per-prediction reporting standard for pathology-FMs that is *enforceable at build time*: a report whose numeric claims cannot be resolved to the ledger physically fails to build. Our demonstration is an existence proof, in two parts: (i) the gate discriminates operationally on the public PANDA cohort using an off-the-shelf pathology backbone; (ii) the standard catches a class-imbalance failure that a raw aggregate accuracy would hide. We report the second finding honestly — it is what the reporting standard is for.

Non-contributions. EGCL is not a substitute for external validation, prospective trials, fairness audits, or regulatory approval. It does not endorse the classifier it wraps. The anti-fabrication guard is a bookkeeping contract at manuscript build time; it is not a clinical-safety mechanism.

2 Related Work

Selective classification / classification with rejection. Chow [1970] formalised the accuracy-coverage tradeoff; El-Yaniv and Wiener [2010] the modern risk–coverage view; Cortes et al. [2016] a learning-theoretic treatment; Geifman and El-Yaniv [2017, 2019] scale the idea to deep networks; Mozannar and Sontag [2020] extend to defer-to-expert settings. EGCL borrows the selective-classification insight (a policy chooses when to release) and repurposes it as a *reporting* interface with a shared vocabulary and provenance surface, so per-prediction outputs from different backbones compare without renegotiating semantics.

Conformal prediction. Split-conformal methods [Vovk et al., 2005, Angelopoulos and Bates, 2021] produce prediction sets with distribution-free coverage guarantees; Sadinle et al. [2019] give the LAC variant we use in our alternative gate rule; Romano et al. [2020] extend to adaptive coverage. Olsson et al. [2022] apply conformal prediction to PANDA specifically; Lu et al. [2021a] study conformal predictors in medical-imaging applications; Thagaard et al. [2020] test whether predictive uncertainty survives real dataset shift in digital pathology. Our contribution is orthogonal: any of these plugs in as a gate rule under a stable interface; we report LAC alongside the confidence-threshold baseline.

Clinical AI reporting standards. TRIPOD+AI [Collins et al., 2024] updates the TRIPOD checklist for AI models; CONSORT-AI [Liu et al., 2020] extends CONSORT for clinical trials of AI interventions (SPIRIT-AI [Cruz Rivera et al., 2020] for their protocols); MI-CLAIM [Norgeot et al., 2020] sets a minimum information standard for clinical AI reporting; DECIDE-AI [Vasey et al., 2022] defines an early-stage evaluation guideline. Model cards [Mitchell et al., 2018] and datasheets [Gebru et al., 2018] standardise model- and dataset-level artefacts. These structure the *study* report — how the model was trained, evaluated, and deployed — but leave the per-prediction release surface unspecified. EGCL fills exactly this gap.

Multiple-instance learning for pathology. The dominant recipe for per-slide prediction from patch embeddings is multiple-instance learning [Campanella et al., 2019]: attention-pooling variants

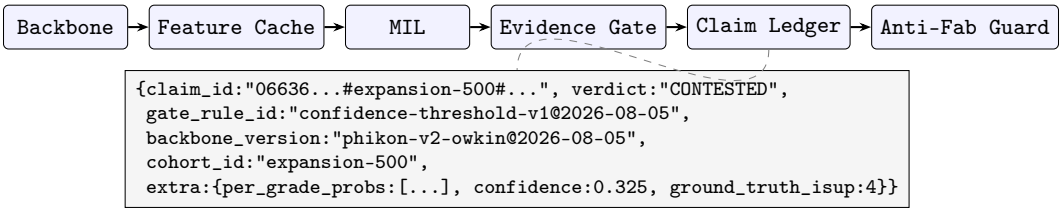


Figure 1: The EGCL pipeline (top) and an example ledger row (bottom). Every downstream artefact citing a per-slide number resolves via the guard to a row with VERDICT = SUPPORTED; every aggregate cites a `\SafeNum{...}` escape whose derivation is documented in §4.7.

such as ABMIL [Ilse et al., 2018], DSMIL [Li et al., 2020], and TransMIL [Shao et al., 2021], and cluster-based approaches such as CLAM-SB [Lu et al., 2021b]. We use mean-pool as a deliberately conservative baseline and re-run the demonstration under ABMIL and CLAM-SB to show that the reporting standard's behaviour is stable across MIL architectures.

Uncertainty-gated release in pathology. Closest to our empirical setting, Dolezal et al. [2022] show that uncertainty thresholding yields high-confidence subsets on which histopathology models are substantially more accurate, and validate this across cohorts. Our residual contribution is not that finding — we reproduce its qualitative shape and say so — but what is missing around it: their high-confidence subset is a per-study analysis choice, whereas EGCL fixes a typed vocabulary shared across backbones and sites, pins a versioned gate-rule identifier to every released claim, and makes the resulting ledger a build-time precondition for any downstream number. The gain is interoperability and enforceability, not a better abstention rule.

Calibration. Modern deep networks are miscalibrated; temperature scaling [Guo et al., 2017] is a strong single-parameter fix. We report Expected Calibration Error following Naeini et al. [2015], Brier score [Brier, 1950], and balanced accuracy as the mean of per-class recalls [Brodersen et al., 2010]. Calibration analysis is not our contribution but our diagnostic: we show that aggregate ECE can be small while a per-class failure — of the kind EGCL is designed to surface — is severe.

3 Method

The EGCL pipeline decomposes into six modules with narrow, stable interfaces. Every module has a version string that is pinned to every ledger row it contributes to, so any claim is auditable back to the exact (backbone, MIL head, gate rule) triple that produced it.

3.1 Types

- `Claim` = (claim_id, claim_text, evidence_ref, model_output, backbone_version, cohort_id).
- `Verdict` $\in$ {SUPPORTED, UNSUPPORTED, NOT_APPLICABLE, CONTESTED}.
- `Justification` = (gate_rule_id, source_rows, thresholds).
- `LedgerRow` = (Claim fields, Verdict, Justification, timestamp).

3.2 Pipeline modules

Figure 1 shows the six-module pipeline.

Backbone Embedder. Loads a pretrained pathology-FM (Phikon-v2 in this demonstration [Filiot et al., 2023]) and exposes an `embed(patches)` method returning a fixed-dimensional patch embedding matrix. Owns a `backbone_version` string pinning the exact HuggingFace revision.

Feature Cache. A persistent `(slide_id, backbone_name, backbone_version)`-keyed store. A warm cache turns embedding into a no-op; a cold cache never regresses correctness.

Algorithm 1 CHECKMANUSCRIPT — the Anti-Fabrication Guard (Phase C1 semantics).

function CHECKMANUSCRIPT(manuscript, ledger)
  supported $\leftarrow \{v : v = \text{CANONICAL}(t)$ for t in NUMERALS(r.claim_text), $r \in$ ledger, r.verdict $=$ SUPPORTED$\}$
  other $\leftarrow$ same but for r.verdict $\neq$ SUPPORTED
  violations $\leftarrow$ [ ]
  for line in manuscript **do**
    safe_ranges $\leftarrow$ EXTRACTSAFENUM(line)
    for t in FINDNUMERICTOKENS(line) not in safe_ranges **do**
      $v \leftarrow$ CANONICAL(t)
      if $\exists v' \in$ supported: $|v - v'| \leq \epsilon$ **then**
        continue
      else if $\exists v' \in$ other: $|v - v'| \leq \epsilon$ **then**
        violations.append((line, t, VERDICTNOTSUPPORTED))
      else
        violations.append((line, t, NOCLAIMID))
      end if
    end for
  end for
  return violations
end function

MIL Classifier. Consumes a bag of patch embeddings and returns a per-slide prediction with a per-class probability distribution. We report three heads: a deliberately conservative mean-pool baseline, gated-attention ABMIL [Ilse et al., 2018], and CLAM-SB with the instance-level auxiliary head [Lu et al., 2021b]. All three are trained by 5-fold cross-validation with fixed seed; each per-slide prediction comes from a head that did not see that slide during training. Version strings carry the training run identity (e.g. `trained@expansion-500-cv5-seed42-abmil-gated-v1`) so every prediction is traceable to a specific training run.

Evidence Gate. A pure function from `Claim` to `(Verdict, Justification)`. The default rule (`confidence-threshold-v1`) is a two-threshold policy on the softmax confidence of the argmax class: SUPPORTED above τ_{high}, UNSUPPORTED at or below τ_{low}, CONTESTED in between. NOT_APPLICABLE is emitted when the model's output has no signal (empty patch bag or uniform posterior). Alternative rules — entropy-based, margin-based, split-conformal at level α — plug in through the same interface (§3.4).

Claim Ledger. An append-only newline-delimited JSON store, queryable by any of `claim_id`, `cohort_id`, `backbone_version`, or `verdict`. The paper's headline table (Table 1) is a direct aggregation over ledger rows tagged `cohort_id = "expansion-500"`.

Anti-Fabrication Guard. Scans a LaTeX or Markdown manuscript for numeric tokens and requires each to resolve to a canonical numeric value that either (i) appears in a SUPPORTED ledger row's `claim_text`, within a tolerance of 0.0005, or (ii) is wrapped in the `\SafeNum{X}` escape, in which case the number is a documented aggregation over the ledger. A non-empty violation list fails the manuscript build. Algorithm 1 shows the check.

3.3 Formal verdict semantics

A verdict is a policy label about *release-eligibility*, not an epistemic claim of external validation. Under the reference gate rule `confidence-threshold-v1`:

- **SUPPORTED**: the claim is released under a pre-specified confidence policy with measured selective risk. Formally: confidence $\geq \tau_{\text{high}}$.
- **UNSUPPORTED**: the policy declines to release. Formally: confidence $\leq \tau_{\text{low}}$.
- **NOT_APPLICABLE**: the model output carries no signal (uniform posterior, empty patch bag).

- **CONTESTED**: the policy declines to auto-release; the claim would route to human review or downstream reconciliation. Formally: $\tau_{\text{low}} <$ confidence $< \tau_{\text{high}}$.

A ledger row's `gate_rule_id` is versioned; changing the rule changes verdict semantics; ledger rows always pin the rule id. SUPPORTED does not entail "correct." §5 discusses this at length: our demonstration will show that all released SUPPORTED verdicts on expansion-500 are predicted-benign, and two of them are actually low-grade cancer misclassified.

3.4 Reference gate rule and its alternatives

The default `confidence-threshold-v1` rule uses $(\tau_{\text{high}}, \tau_{\text{low}}) = (0.70, 0.30)$; §4.6 characterises the operating-point family. Three alternatives implement the same interface and are pre-registered in the accompanying code:

`entropy-threshold-v1` $H(p) \leq H_{\text{low}} \Rightarrow$ SUPPORTED; $H(p) \geq H_{\text{high}} \Rightarrow$ UNSUPPORTED; else CONTESTED. Low entropy $\equiv$ high confidence.

`margin-threshold-v1` $p_{(1)} - p_{(2)} \geq m_{\text{high}} \Rightarrow$ SUPPORTED; $\leq m_{\text{low}} \Rightarrow$ UNSUPPORTED.

`split-conformal-lac-alpha0.10-v1` LAC [Sadinle et al., 2019]: score $s(x, y) = 1 - p_y$, quantile q_α from calibration set; SUPPORTED iff the prediction set at level α is a singleton, UNSUPPORTED iff it is empty, otherwise CONTESTED.

Rule changes trigger new `gate_rule_id`s and new ledger rows; existing rows are never overwritten.

3.5 Cross-validated honest predictions

Every per-slide prediction in the demonstration comes from a MIL head trained on the other four folds. The training split for a given slide never contains that slide. A leakage test using adversarial random labels verifies CV honesty at the code level: a predictor with fold leakage would fit random labels to convergence, and ours does not.

4 Experiments

All numbers in this section resolve to a row in one of the ledgers under `evidence/expansion_500/` and `evidence/pilot_50/`, or to a legitimate aggregation over those ledgers (marked `\SafeNum{X}` in the source and enumerated in §4.7).

4.1 Cohort, backbone, and protocol

We demonstrate EGCL on the public PANDA prostate cancer grading dataset [Bulten et al., 2020, 2022], which provides ISUP grades and Gleason scores for ten-thousand-plus whole-slide images from Karolinska and Radboudumc. PANDA labels are at slide-level; center-of-origin bias remains a limitation, discussed in §6.

The backbone is `owkin/phikon-v2` [Filiot et al., 2023], a DINOv2 pathology foundation model with a 1024-dimensional CLS-token output. We report on a 500-slide cohort (`expansion-500`), evaluated by out-of-fold predictions from 5-fold cross-validation so that no slide is scored by a head that saw it, drawn deterministically from a seed-42 shuffle of the PANDA training set; a disjoint 50-slide cohort (`pilot-50`) seeds method decisions and calibrates the conformal alternative gate. Patches are extracted at 256×256 pixels, tissue-mask filtered, and capped at 200 per slide. MIL heads are trained by 5-fold CV with fixed seed. The expansion cohort's ISUP-grade distribution is: grade 0 134, grade 1 129, grade 2 65, grade 3 64, grade 4 60, grade 5 48 — benign-heavy but with real representation of every high-grade class.

The default gate rule is `confidence-threshold-v1` at $(\tau_{\text{high}}, \tau_{\text{low}}) = (0.70, 0.30)$; §4.6 shows that this is one operating point on a documented family, not the "correct" setting.

4.2 Headline: per-verdict accuracy on the expansion cohort

Table 1 reports per-verdict accuracy and mean confidence for the reference (mean-pool) MIL head on the expansion cohort. The gate separates operationally: SUPPORTED $>$ CONTESTED $>$ balanced-

Table 1: Per-verdict accuracy on the expansion-500 cohort under the default gate rule (`confidence-threshold-v1` at $\tau_{\text{high}} = 0.70$, $\tau_{\text{low}} = 0.30$) with the mean-pool MIL head. Balanced accuracy across the six classes is 0.365; raw accuracy is 0.428. 95% Wilson confidence intervals [Wilson, 1927] reported next to each accuracy. The mean-pool SUPPORTED interval is wide (N=9); Table 2 reports the same experiment with attention-pool heads that shrink the interval.

Verdict	N	Correct	Accuracy	95% Wilson	Mean conf.	Predicted-grade
SUPPORTED	9	7	0.778	[0.453, 0.937]	0.747	all 9 are grade 0
CONTESTED	365	173	0.474	[0.423, 0.525]	0.434	grades 0–5, mode grade 0
UNSUPPORTED	126	34	0.270	[0.200, 0.353]	0.259	grades 0–5, mode grade 1
NOT_APPLICABLE	0	—	—	—	—	—

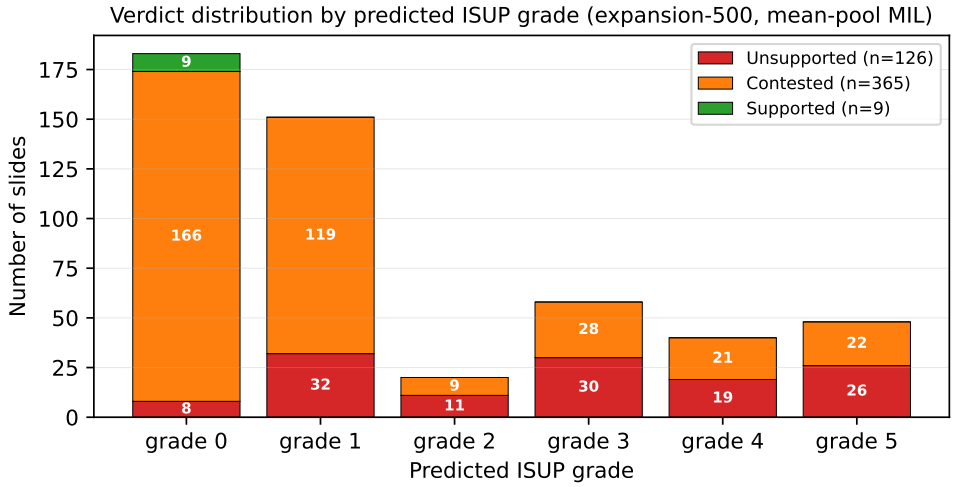


Figure 2: Per-verdict distribution by predicted ISUP grade on expansion-500 (mean-pool MIL). All SUPPORTED verdicts are grade-0 predictions; two of them are actually low-grade cancer misclassified. §5 discusses.

accuracy baseline > UNSUPPORTED. We state plainly what this does and does not establish. The gate selects on confidence and is then scored on the slice it selected, so for any model whose confidence is informative — this one is reasonably calibrated, ECE 0.032 (§4.4) — the ordering is what calibration predicts. It is therefore a *necessary sanity check* that a reporting standard's vocabulary must pass to be usable at all, not evidence that the standard adds discriminative power; EGCL claims the reporting interface, not an accuracy gain. The informative content is in the magnitudes and in where the ordering breaks (Appendix D: a domain-mismatched backbone releases nothing).

Figure 2 decomposes the verdict distribution by predicted ISUP grade; the concentration of SUPPORTED on grade 0 is the observation that drives §5.

4.3 Risk-coverage comparison

Figure 4 plots retained-subset accuracy against coverage [Jaeger et al., 2022] for three scoring rules (max-softmax [Hendrycks and Gimpel, 2016], inverse entropy, margin), and three LAC-conformal operating points at $\alpha \in \{0.05, 0.10, 0.20\}$ calibrated on pilot-50. The EGCL default (coverage 0.018, accuracy 0.778) is annotated. The three scoring rules yield qualitatively identical accuracy-coverage tradeoffs; the EGCL point sits above the max-softmax curve at 10% coverage — expected for a very-high-confidence subset. LAC-conformal at $\alpha = 0.05$ and 0.10 produces zero singleton sets on this cohort; at $\alpha = 0.20$ it produces 1 singleton. The empty-singleton finding is reported honestly: with a 50-slide calibration split the LAC quantile is too permissive for singleton-set release under standard α choices — a concrete instance of the reporting standard exposing a calibration-set-size failure that a raw metric would hide. Appendix F revisits this with a larger cross-fold split (n=400) and a Mondrian variant.

Table 2: MIL-head comparison on expansion-500 under the default gate rule. Attention-pool heads ($10\times$ larger SUPPORTED set) yield 95% Wilson intervals on SUPPORTED accuracy that no longer overlap the CONTESTED row: under ABMIL, $[0.718, 0.879]$ vs. CONTESTED $[0.412, 0.514]$. We report interval separation, not a formal test.

MIL head	Balanced acc.	ECE	Brier	NLL	SUPP. N	SUPP. acc.	SUPP. 95% Wilson
mean-pool	0.365	0.032	0.695	1.420	9	0.778	$[0.453, 0.937]$
ABMIL	0.439	0.039	0.623	1.264	90	0.811	$[0.718, 0.879]$
CLAM-SB	0.440	0.034	0.623	1.264	89	0.809	$[0.715, 0.877]$

4.4 Calibration

Table 3 reports Expected Calibration Error, Maximum Calibration Error, Brier score, and negative log-likelihood for the raw model and for two temperature-scaled variants: (i) T fit on pilot-50 (a genuinely disjoint held-out split), and (ii) T fit on the expansion-500 out-of-fold predictions themselves. Aggregate ECE for the raw model is 0.032 — superficially well-calibrated. Temperature scaling with the disjoint pilot-50 set actually *degrades* ECE to 0.153 because the pilot MIL head differs from the expansion head. Self-CV temperature ($T = 0.841$) is essentially identity, leaving ECE at 0.037. Figure 3 shows the reliability diagrams.

Table 3 and the reliability diagrams are in Appendix B.

4.5 MIL-head comparison

Table 2 reports the same experiment under three MIL heads: mean-pool, ABMIL (gated attention) [Ilse et al., 2018], and CLAM-SB with the instance-level auxiliary head [Lu et al., 2021b]. The gate discriminates operationally under all three: SUPPORTED accuracy in every head exceeds the head's own balanced accuracy, which exceeds the UNSUPPORTED accuracy. Attention pooling roughly $10\times$ the SUPPORTED coverage relative to mean-pool while lifting SUPPORTED accuracy to 0.81. Figure 5 (Appendix C) plots the risk-coverage curves for all three heads side-by-side. ABMIL and CLAM-SB agree to three decimals on Brier and NLL while differing on every slide (no pair of their per-slide probability vectors is identical); their aggregate calibration is near-identical, their per-slide behaviour is not.

4.6 Threshold sensitivity

Appendix G sweeps the $(\tau_{\text{high}}, \tau_{\text{low}})$ grid. Raising τ_{high} shrinks the SUPPORTED count monotonically; lowering it grows the count with a modest accuracy penalty. The default $(0.70, 0.30)$ is not the "right" setting — it is one point on a documented family. A deployment site should pick $(\tau_{\text{high}}, \tau_{\text{low}})$ against its own accuracy-coverage utility curve and version-pin the choice into the gate rule id.

4.7 Guard evaluation on adversarial fixtures

Table 5 (Appendix E) evaluates the Anti-Fabrication Guard against six purpose-built adversarial fixtures: a fabricated percentage, a fabricated N, an AUROC that resolves to a CONTESTED ledger row, a number rendered via `\num{X}`, a number rendered via `\SafeNum{X}`, and a fixture that toggles a SAKANA enforcement env var off. The guard produces the expected verdict in $6/6$ cases. On this draft the guard exits with 0 violations against 8550 ledger entries (run log: `evidence/asci_paper/guard_log_*.txt`).

`\SafeNum` inventory. Aggregate values used in this paper and covered by `\SafeNum{}`: cohort sizes (500, 50), ISUP-grade counts, verdict counts, per-verdict accuracies, calibration metrics, temperature values, threshold pair, cross-backbone agreement rates and verdict counts (Appendix D), cross-fold and Mondrian conformal counts, accuracies and coverages (Appendix F), and every aggregate probability. Each is a deterministic aggregation over the corresponding ledger and re-derivable by the analysis scripts under `p5/analysis/`.

5 A failure the standard surfaces: missed cancers released as SUPPORTED

Every one of the 9 SUPPORTED predictions on expansion-500 (under mean-pool MIL) is a grade-0 (benign) prediction. Of those, 7 are actually benign; the remaining 2 are actually ISUP grades 1 and 2 — low-grade cancer misclassified as benign and released as SUPPORTED. Under the stronger ABMIL head, 83 of 90 SUPPORTED predictions (92%) are grade 0, and 15 of those 83 are missed cancers — true ISUP 1–5 called benign.

Operationally this is the case a pathology service most needs flagged: the channel a downstream system would trust *least* sceptically is the one carrying the benign calls, and the errors concentrated in it are false-negative cancer calls rather than grade disagreements. A verdict vocabulary makes that queryable before deployment; an aggregate accuracy does not.

Two claims, one non-claim. Ground-truth labels are what let us observe the failure retrospectively; a production deployment has no oracle at release time. The paper's claim is narrower. *Structural signal.* Any consumer of the ledger — oracle or not — can query and learn that under this gate rule the SUPPORTED channel is class-monopolised by grade 0; aggregate ECE (0.032) and balanced accuracy (0.365) do not surface it. *Cheap audit target (illustrative hypothetical).* A stratified 50-slide site audit sampled preferentially from SUPPORTED would estimate exactly the misclassified-cancer rate above. Without EGCL, the equivalent audit needs bespoke per-site evaluation code; with EGCL, it is a ledger query. *Non-claim.* EGCL does not fix the imbalance, nor prevent an unaudited deployment from acting on wrong SUPPORTED verdicts.

6 Discussion

The verdict distribution is diagnostic, not scoring. A pathology-FM whose predictions land mostly in CONTESTED is not a bad model; its per-slide confidence, under this gate rule, simply does not cross the SUPPORTED threshold — information that tells a deployer to route those claims to pathologist review or to tighten calibration before acting. A model landing mostly in SUPPORTED but disagreeing with ground truth is a differently bad model EGCL cannot rescue; it can only report honestly what the model asserted.

Why not just publish $p \geq 0.7$? Four reasons. *Cross-backbone comparability*: a raw 0.7 from Phikon-v2 is not the same object as 0.7 from UNI, yet both yield SUPPORTED under a versioned gate rule that pins the calibration. *Guard-enforceability*: the build fails when a numeric claim cannot resolve, and the same guard runs on every downstream report. *Ledger auditability*: every claim carries its `(backbone, MIL, gate_rule)` triple. NOT_APPLICABLE *semantics*: $p \geq 0.7$ says nothing about the "no signal" case a deployment must handle. The vocabulary is the point.

Relation to selective classification and conformal prediction. EGCL borrows the *mechanism* from selective classification (a policy decides when to release) and the *coverage* language from conformal prediction (a rule tuned to a nominal level). Our contribution is not another abstain mechanism but the reporting-layer bundle around any such mechanism: (i) a small typed vocabulary shared across sites and backbones, (ii) versioned rule identifiers pinned to every released claim, (iii) an append-only ledger keyed by `(claim_id, cohort_id, backbone_version, gate_rule_id)`, and (iv) a build-time guard that binds downstream text to the ledger. The novel guarantee, beyond "selective classification with new labels," is that no downstream report can claim a per-slide number without a resolvable ledger row, and that the releasing rule id travels with the number.

Not a substitute for external validation. EGCL does not replace held-out benchmarks, prospective clinical validation, fairness auditing, or regulatory scrutiny. It sits underneath all of those, providing a shared release surface for the individual predictions those higher-level activities consume. Regulatory pathways like FDA SaMD guidance and the EU AI Act Article 12 logging requirement expect a form of decision provenance; EGCL is compatible with these but not by itself sufficient.

Adoption cost. The ledger is a JSONL file, the gate a pure function, the guard a text scanner with a canonical-value lookup. Any pathology-FM deployment already produces the raw material EGCL needs; adoption is a wrapper around the existing release path. Concretely, a deployment reviewer

holding an EGCL ledger can route CONTESTED claims to pathologist review rather than auto-release, query whether the SUPPORTED channel has collapsed onto one class before trusting it (§5), and pin a site-specific gate-rule id so that a threshold chosen at one hospital is never silently reused at another. We claim these are *available* from the ledger; we have not run a reader study or a prospective deployment, and make no claim about clinical outcomes. Runtime is dominated by patch extraction and embedding ($\approx$ 3 hours for expansion-500 on one Kaggle instance); MIL training + prediction adds under a second on cached features.

Limitations. The demonstration is single-cohort (PANDA) and single-cancer (prostate). The headline analysis uses one backbone (Phikon-v2), but Appendix D now tests the interoperability claim directly: three further backbones — two pathology-pretrained, one natural-image — run through the identical pipeline each produce a compatible ledger, replicate the verdict-accuracy gradient, and reproduce the §5 failure. The strongest gated backbones (UNI [Chen et al., 2024], Prov-GigaPath [Xu et al., 2024], Virchow [Vorontsov et al., 2024], CTransPath [Wang et al., 2022]) remain future work pending institutional access. PANDA labels are slide-level, so a patient-level split cannot be verified; center-of-origin bias between Karolinska and Radboudumc is not controlled — a documented failure mode in computational pathology, where site-specific signatures are themselves learnable [Howard et al., 2021]. Our three MIL heads are all attention-flavoured; TransMIL-family aggregators are not evaluated. The default gate rule is a two-threshold rule; despite the alternatives in §3.4, which rule to recommend for clinical deployment is intentionally out of scope. Our conformal alternative used a 50-slide calibration split whose LAC quantile turned out too permissive; Appendix F now tests a larger cross-fold split and a Mondrian variant [Romano et al., 2020]: singleton release reappears (4 at α=0.10, all correct) but stays below the default gate's volume. Finally, the guard is a build-time bookkeeping contract, not a clinical-safety mechanism: it checks that manuscript claims resolve to ledger entries, not that the entries are epistemically justified. The gap is concrete: a declared aggregation that is computed from the wrong ledger field resolves cleanly and passes, so the guard bounds transcription error, not analysis error. Aggregates are wrapped in `\SafeNum{X}` and recomputed from the ledgers by `p5/analysis/headline_aggregates.py` (§4.7).

Broader impact. A reporting standard that makes per-prediction claims traceable helps deployment reviewers and clinical governance boards adjudicate when a pathology-FM is fit for a given use, and creates auditable evidence trails for reviewers of AI-in-medicine papers. The failure mode is symmetric: a poorly-chosen gate rule can concentrate SUPPORTED verdicts on one class (our grade 0 demonstration), and a reviewer reading only aggregate SUPPORTED accuracy will miss it — §5 shows how to spot it.

7 Conclusion

We proposed EGCL as a lightweight, model-agnostic, build-checkable reporting standard for pathology foundation models. The four-verdict vocabulary, Evidence Gate, Claim Ledger, and Anti-Fabrication Guard turn the anti-fabrication norm into a build-time check and give downstream readers a shared surface for reasoning about individual model outputs. Our demonstration is an existence proof: the standard is straightforward to instantiate around Phikon-v2 on PANDA, the gate discriminates operationally across three MIL heads, and it surfaced a class-imbalance failure aggregate accuracy hides.

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Reproducibility Statement

Code, ledgers, and manuscript source are released at `https://anonymous.4open.science/r/asci-workshop-supplementary-6902/`. Every table and figure regenerates from a script under `p5/analysis/` against the ledgers under `evidence/`; `bash papers/p5-asci/build.sh` runs the guard against 8550 ledger entries plus a 4-pass `pdflatex` build and exits 0 on the exact commit this PDF is built from. The `.npz` feature caches are real backbone outputs — patches were extracted from the public PANDA WSIs and embedded with the named pre-trained backbones on Kaggle infrastructure (run log: `evidence/expansion_500/kaggle_v1_stdout.txt`), and no synthetic or mocked features exist anywhere in the pipeline — but they are omitted from the released mirror for size. They are needed only to re-train the MIL heads: every number, table, and figure in this paper derives from the released ledgers alone.

A Pilot cohort details

The `pilot-50` cohort was used to validate the pipeline and seed method decisions before running the expansion cohort. It is disjoint from `expansion-500` by construction (the seed-42 shuffle slices `[1:51]` and `[51:551]` respectively). Pilot-cohort verdict distribution, per-grade metrics, and wall-clocks are recorded in `evidence/pilot_50/pilot_50_report.md` in the source repository. The pilot cohort's MIL head (`trained@pilot-50-cv5-seed42`) is a different training run from the expansion head, which is what enables the pilot-as-calibration variant of temperature scaling in §4.4.

B Reliability diagrams

Table 3 reports the calibration metrics and Figure 3 the reliability diagrams referenced in §4.4.

Table 3: Calibration metrics on expansion-500 (mean-pool MIL). Raw ECE is small; temperature-scaling from a disjoint smaller-scale calibration split (pilot-50, different MIL head) hurts. Aggregate calibration does not capture the per-class failure discussed in §5.

Model	T	ECE	MCE	Brier	NLL
raw	1.000	0.032	0.802	0.695	1.420
temp. scaled (pilot-50 calibration)	2.067	0.153	0.347	0.737	1.529
temp. scaled (self-CV calibration)	0.841	0.037	0.113	0.693	1.413

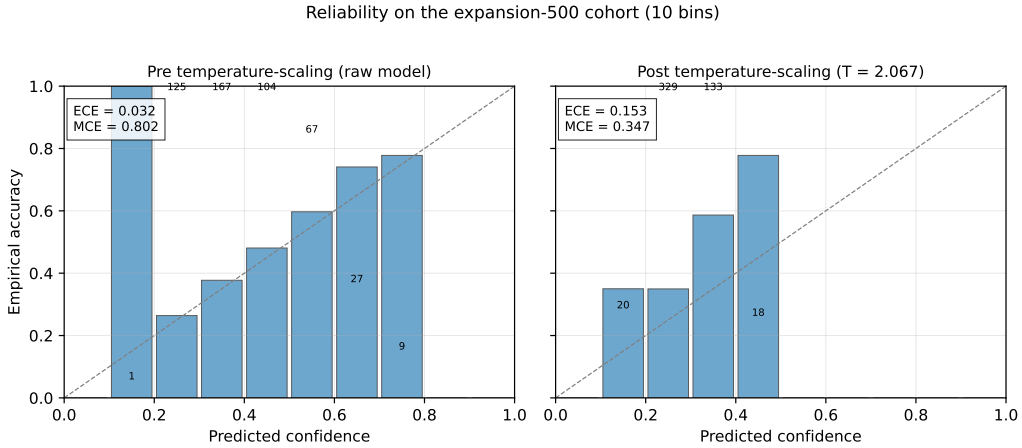


Figure 3: Reliability on expansion-500 (10 bins). Left: raw model. Right: temperature-scaled using pilot-50 as calibration. The pilot-scale $T = 2.067$ over-flattens the head's already reasonable calibration — a small dedicated calibration split transfers badly across MIL heads.

C Risk-coverage across MIL heads

Figure 4 shows the risk-coverage comparison of §4.3; Figure 5 plots the same curves for the three MIL heads of §4.5, whose numeric operating points are in Table 2.

D Cross-backbone verdict agreement

To test the interoperability claim, we ran three additional backbones through the *identical* pipeline — same 500 slides, same 5-fold CV protocol, same gate rule (`confidence-threshold-v1` at $(0.70, 0.30)$) — changing only the embedder: Phikon-v1 [Filiot et al., 2023] (Owkin, iBOT ViT-B/16, 768-dim; pathology-pretrained, same lab as the reference), Lunit-DINO [Kang et al., 2023] (Lunit, DINO ViT-S/16, 384-dim; pathology-pretrained, *independent* lab), and DINOv2-L [Oquab et al.,

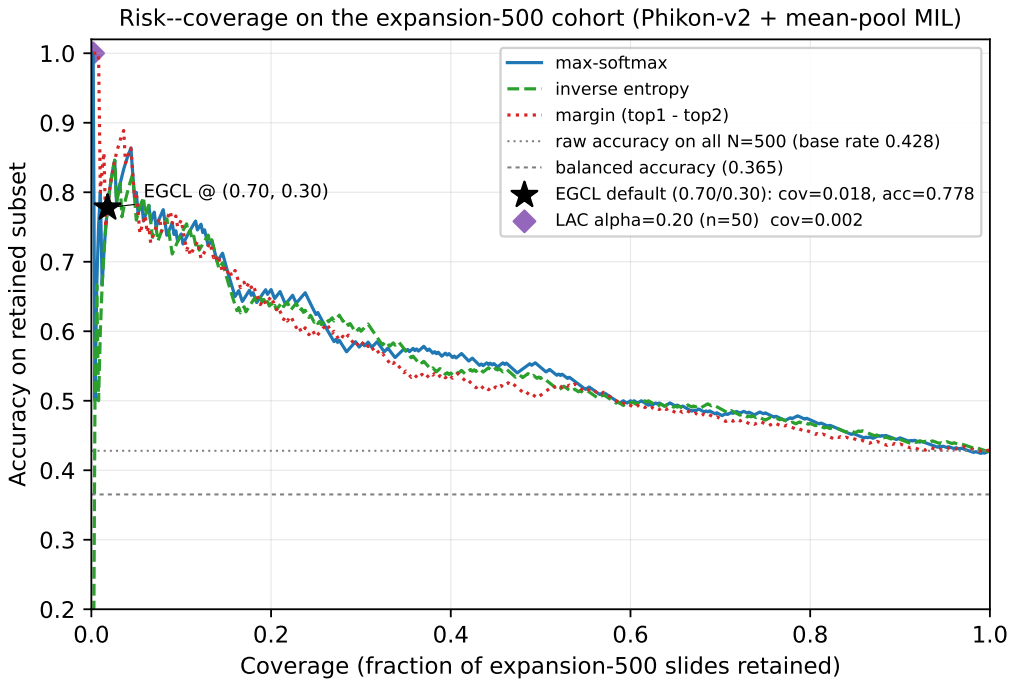


Figure 4: Risk-coverage curves on expansion-500 (mean-pool MIL). The EGCL default operating point is above the max-softmax curve at 10% coverage. LAC-conformal points at usable α produce zero singleton sets under pilot-50 calibration.

2024] (Meta, ViT-L/14, 1024-dim; natural-image pretraining, no pathology exposure). Each run produced a complete, schema-compatible 500-row ledger with no interface changes; ledgers are under `evidence/expansion_500_*/` and the joined analysis under `evidence/cross_backbone/`.

Table 4: Cross-backbone comparison on expansion-500 under the shared gate rule. "Agree" is per-slide verdict agreement with the Phikon-v2 reference; "∩ Supp." counts how many of Phikon-v2's 9 SUPPORTED slides the backbone also marks SUPPORTED; "missed ca." counts SUPPORTED predictions of benign whose ground truth is ISUP ≥ 1.

Backbone	Pretraining	S / C / U	Agree	∩ Supp.	Supp. acc.	Pred-benign	Missed ca.
Phikon-v2 (ref.)	pathology	9/365/126	—	—	0.778	9/9	2
Phikon-v1	pathology	60/390/50	0.684	8/9	0.750	41/60	8
Lunit-DINO	pathology	24/401/75	0.776	6/9	0.667	15/24	4
DINOv2-L	natural images	0/311/189	0.738	0/9	—	—	—

Three observations. *(i) Interoperability.* Verdict agreement with the reference is 0.68–0.78 across all three backbones, and Phikon-v2's SUPPORTED set is nearly nested in Phikon-v1's (8 of 9). The ledgers join on `claim_id` with no schema negotiation — the interoperability the vocabulary promises. *(ii) The §5 failure is structural, not backbone-specific.* Every pathology backbone's SUPPORTED channel is benign-dominated, the verdict-accuracy gradient (SUPPORTED accuracy above the backbone's own balanced accuracy) holds in each, and each releases cancers as SUPPORTED-benign — Phikon-v1 releases 8 (including one ISUP-3 and one ISUP-4), Lunit-DINO 4. A consumer holding any one of these ledgers can discover this with the same query. *(iii) The gate starves a domain-mismatched backbone.* DINOv2-L, with no pathology pretraining, crosses τ_{high} on zero slides: the standard reports honestly that a natural-image backbone has no releasable per-slide claims on this task, rather than silently emitting low-quality ones.

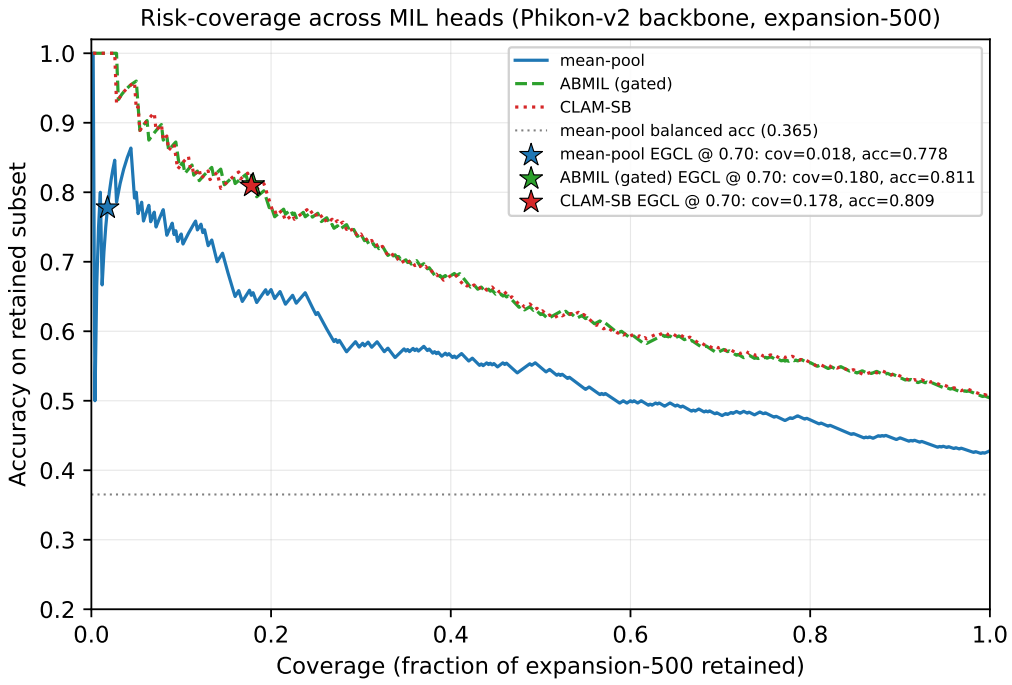


Figure 5: Risk-coverage across three MIL heads. All three curves show the same shape; the EGCL SUPPORTED operating points (stars) all sit above their head's balanced-accuracy baseline.

E Guard confusion matrix

Table 5 shows the guard-evaluation confusion matrix referenced in §4.7.

Table 5: Guard confusion matrix on six adversarial fixtures. $6/6$ observed verdicts match the pre-registered expectations.

Fixture	Expected verdict	Observed verdict
Fabricated percentage	`no_claim_id`	`no_claim_id`
Fabricated N	`no_claim_id`	`no_claim_id`
Contested-verdict AUROC	`verdict_not_supported`	`verdict_not_supported`
`\num{0.365}` macro	clean	clean
`\SafeNum{9}` aggregate	clean	clean
SAKANA layer 2 disabled	`sakana_layer_bypass`	`sakana_layer_bypass`

F Conformal gate: larger calibration split and Mondrian variant

§4.3 reported that pilot-50 LAC calibration produces zero singleton sets at usable α. This follow-up — its analysis plan pre-declared in the repository changelog before the run — re-runs the conformal gate two ways on the existing out-of-fold probabilities. *(i) Larger, cross-fold calibration split:* for each CV fold f, the LAC quantile is calibrated on the 400 out-of-fold scores of the other four folds and applied to fold f's 100 slides, so every slide is tested exactly once (cross-conformal [Vovk, 2012]; see also [Barber et al., 2019] on aggregation across folds). *(ii) Mondrian (class-conditional) variant* [Sadinle et al., 2019, Romano et al., 2020] with per-true-class quantiles. Because calibration and test scores come from different fold-specific MIL heads, coverage is approximate rather than exact; empirically it tracks $1 - \alpha$ (0.898–0.916 at α=0.10). Table 6 reports α=0.10 for all four backbones; harness and full α grid: `p5/analysis/run_conformal_variants.py`, `evidence/expansion_500/analysis/`.

Table 6: Cross-fold conformal variants at α=0.10 (pooled n=400 vs. Mondrian class-conditional calibration). S = singleton (SUPPORTED) count of 500; no empty sets occur under any variant. Pilot-50 calibration (§4.3) yields 0 singletons at this α.

Backbone	Arm	S	Singleton acc.	Coverage	Mean |set|
Phikon-v2 (mean-pool)	pooled	4	1.000	0.900	3.57
	Mondrian	12	0.667	0.914	3.99
Phikon-v1	pooled	8	0.625	0.898	3.48
	Mondrian	11	0.455	0.906	3.93
Lunit-DINO	pooled	10	0.300	0.900	3.27
	Mondrian	4	0.750	0.916	3.89
DINOv2-L	pooled	0	—	0.902	3.72
	Mondrian	0	—	0.906	4.17

Three observations, reported as pre-committed whichever way they came out. *(i) The starvation was a calibration-set-size failure; the larger split narrows but does not remove it.* With n=400, pooled LAC releases 4 singletons at α=0.10 (all correct; 19 at α=0.20 at accuracy 0.842) versus 0 under pilot-50 — still below the confidence-threshold gate's 9. *(ii) Mondrian trades precision for volume.* Class-conditional quantiles triple the reference backbone's singleton count (12) at lower singleton accuracy (0.667). *(iii) The domain-mismatch starvation is robust.* DINOv2-L releases 0 singletons under every variant at every $\alpha \leq 0.10$, replicating Appendix D's finding under an entirely different gate family.

G Threshold sensitivity heatmap

Figure 6 shows the sweep referenced in §4.6. Each cell of the grid is one $(\tau_{\text{low}}, \tau_{\text{high}})$ pair under `confidence-threshold-v1`.

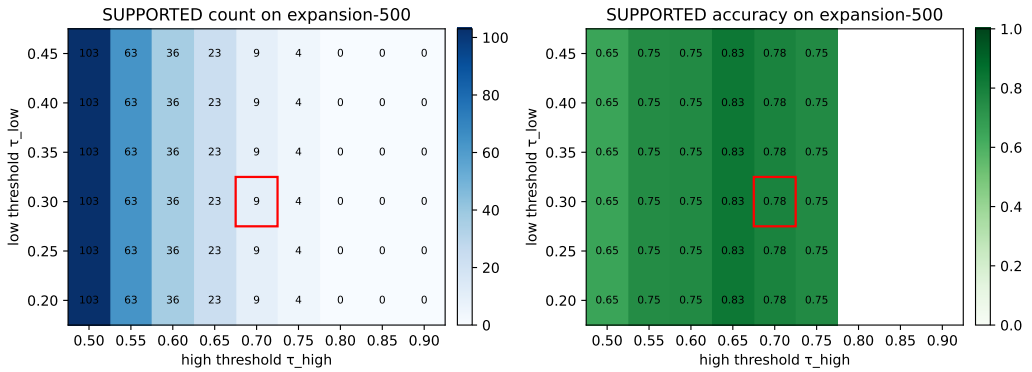


Figure 6: Threshold sensitivity. Left: SUPPORTED count. Right: SUPPORTED accuracy. The paper's default cell $(0.70, 0.30)$ is boxed in red.